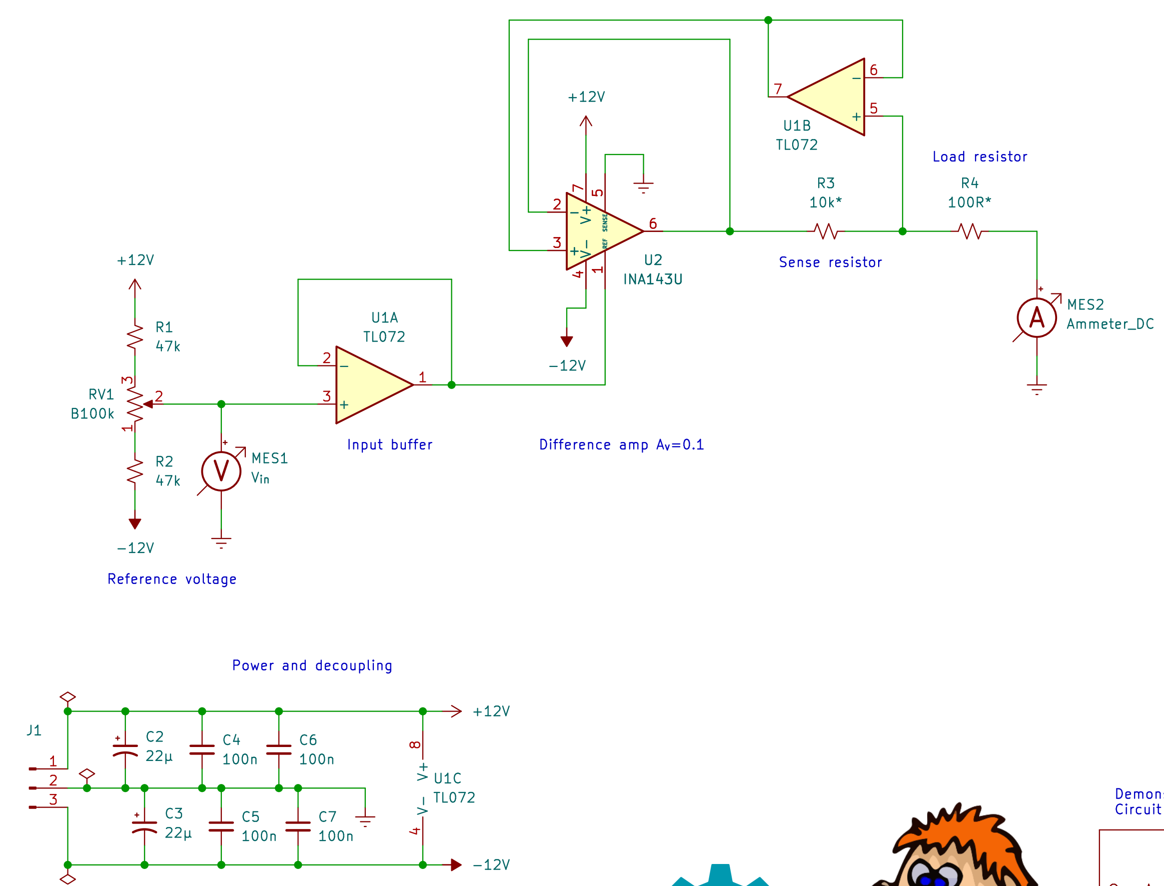

Place one pair of 100 nF capacitors near the power pins of each IC

The change from the first version is the addition of U1B to buffer the feedback signal, so that the low impedance of U2's inputs does not steal current from the load.

WIth that change made, the circuit performs well down to microamp levels of current with a 10k sense resistor.

Changing the sense resistor to 10 ohms, and using 3.3 ohms (three 10 ohm, 5 W resistors in parallel) as the load resistor shows the limitation on output current. U2 cannnot source nor sink more than about 20−30 mA.

All resistors are 1/4 W 1% metal film except as noted.
* − See description for the values used in the different tests.
C2 and C3 are 25V aluminum electrolytic; all others are 50V X5R or Y7V ceramic.

Demonstration repeated with addition of feedback buffer.
Circuit now performs well with microamp−level currents and a 10k sense resistor.

Op−Amp Basics
Episode 17 − Transcoductance Amplifiers 4
Kludges from Kevin's Cave

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Title: Widlar Current Pump

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